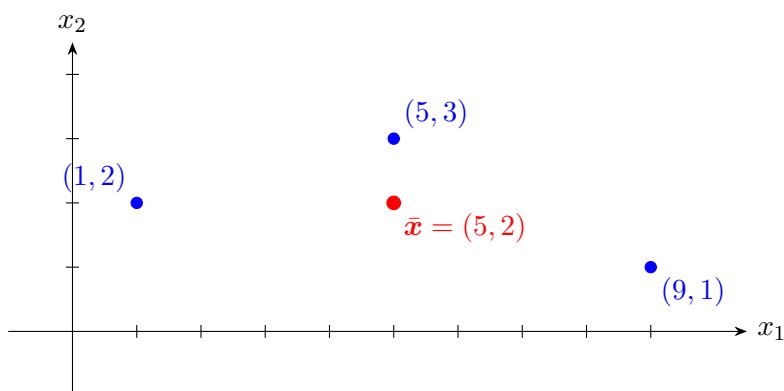


Problem 3.1 (8 points)

Given $X = \begin{pmatrix} 9 & 1 \\ 5 & 3 \\ 1 & 2 \end{pmatrix}$, so $n = 3$, $p = 2$. The sample mean is $\bar{x}_1 = \frac{9+5+1}{3} = 5$, $\bar{x}_2 = \frac{1+3+2}{3} = 2$, so $\bar{\mathbf{x}} = (5, 2)'$.

(a) Scatter plot in $p = 2$ dimensions

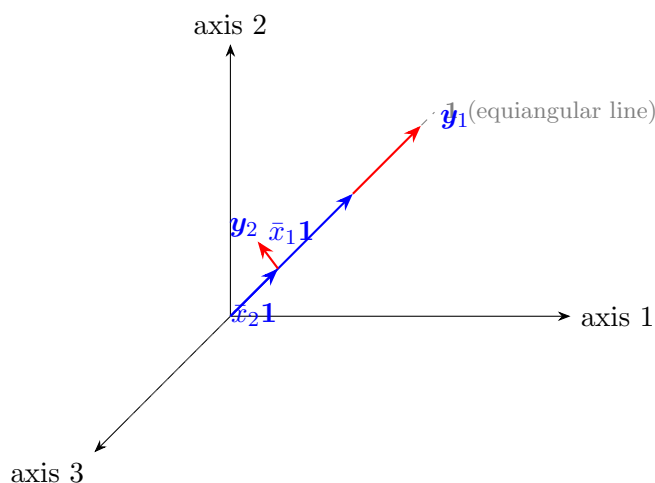
The three rows of X are plotted as points $(9, 1)$, $(5, 3)$, $(1, 2)$ in the (x_1, x_2) -plane, with the mean $\bar{\mathbf{x}} = (5, 2)$ marked as the centroid of the cloud.



(b) $n = 3$ -dimensional representation and deviation vectors

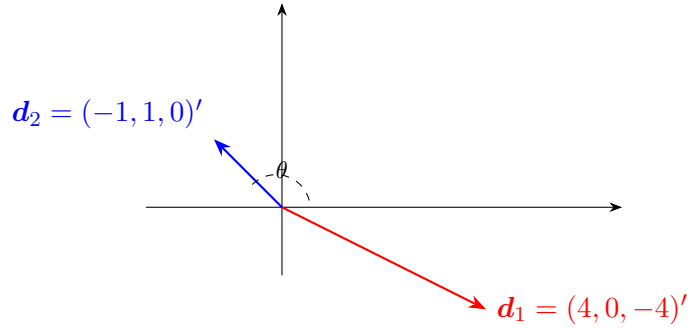
Each column (variable) of X is plotted as a vector in $\mathbb{R}^n = \mathbb{R}^3$: $\mathbf{y}_1 = (9, 5, 1)'$, $\mathbf{y}_2 = (1, 3, 2)'$. The vectors $\bar{x}_1 \mathbf{1} = (5, 5, 5)'$ and $\bar{x}_2 \mathbf{1} = (2, 2, 2)'$ lie on the equiangular line (direction $\mathbf{1} = (1, 1, 1)'$). The deviation vectors are

$$\mathbf{d}_1 = \mathbf{y}_1 - \bar{x}_1 \mathbf{1} = (4, 0, -4)', \quad \mathbf{d}_2 = \mathbf{y}_2 - \bar{x}_2 \mathbf{1} = (-1, 1, 0)'.$$



$\mathbf{y}_1$ and $\mathbf{y}_2$ decompose into a component along the equiangular line, $\bar{x}_j \mathbf{1}$, plus an orthogonal deviation component $\mathbf{d}_j$.

(c) Deviation vectors from the origin, lengths, angle, and relation to S_n and R



Lengths: $\|\mathbf{d}_1\| = \sqrt{4^2 + 0^2 + (-4)^2} = \sqrt{32} = 4\sqrt{2}$, $\|\mathbf{d}_2\| = \sqrt{(-1)^2 + 1^2 + 0^2} = \sqrt{2}$.

Angle: $\mathbf{d}_1 \cdot \mathbf{d}_2 = (4)(-1) + (0)(1) + (-4)(0) = -4$, so

$$\cos \theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{\|\mathbf{d}_1\| \|\mathbf{d}_2\|} = \frac{-4}{(4\sqrt{2})(\sqrt{2})} = \frac{-4}{8} = -\frac{1}{2}, \quad \theta = 120^\circ.$$

Relation to S_n and R : using the divisor- n covariance matrix, $(S_n)_{jk} = \mathbf{d}_j \cdot \mathbf{d}_k / n$, so

$$S_n = \begin{pmatrix} 32/3 & -4/3 \\ -4/3 & 2/3 \end{pmatrix}, \quad r_{12} = \frac{(S_n)_{12}}{\sqrt{(S_n)_{11}(S_n)_{22}}} = \frac{-4/3}{\sqrt{(32/3)(2/3)}} = \frac{-4/3}{8/3} = -\frac{1}{2}.$$

Thus $r_{12} = \cos \theta = -\frac{1}{2}$: the squared length of a deviation vector divided by n gives that variable's sample variance, and the cosine of the angle between two deviation vectors equals the sample correlation between the corresponding variables. The obtuse angle (120°) reflects the negative association between x_1 and x_2 .

Problem 3.6 (8 points)

Given $X = \begin{pmatrix} -1 & 3 & -2 \\ 2 & 4 & 2 \\ 5 & 2 & 3 \end{pmatrix}$, $n = 3$, $p = 3$. Means: $\bar{x}_1 = \frac{-1+2+5}{3} = 2$, $\bar{x}_2 = \frac{3+4+2}{3} = 3$, $\bar{x}_3 = \frac{-2+2+3}{3} = 1$, so $\bar{\mathbf{x}}' = (2, 3, 1)$.

(a) Deviation matrix and rank

$$X - \mathbf{1}\bar{\mathbf{x}}' = \begin{pmatrix} -1-2 & 3-3 & -2-1 \\ 2-2 & 4-3 & 2-1 \\ 5-2 & 2-3 & 3-1 \end{pmatrix} = \begin{pmatrix} -3 & 0 & -3 \\ 0 & 1 & 1 \\ 3 & -1 & 2 \end{pmatrix} = D.$$

Since mean-centering forces $\mathbf{1}'D = \mathbf{1}'X - n\bar{\mathbf{x}}' = \mathbf{0}'$, the three rows of D (as vectors in $\mathbb{R}^3$) must sum to $\mathbf{0}$: indeed, $\text{row}_1 + \text{row}_2 + \text{row}_3 = (-3 + 0 + 3, 0 + 1 - 1, -3 + 1 + 2) = (0, 0, 0)$. This is a nontrivial linear dependence among the rows, so $\text{rank}(D) \leq 2 < 3$. Since $\text{row}_1 = (-3, 0, -3)$ and $\text{row}_2 = (0, 1, 1)$ are not proportional, $\text{rank}(D) = 2$ exactly. **D is not of full rank** — this is a general fact: the mean-corrected data matrix always satisfies $\text{rank}(X - \mathbf{1}\bar{\mathbf{x}}') \leq n - 1$.

(b) Sample covariance matrix S and generalized variance $|S|$

With columns of D given by $\mathbf{d}_1 = (-3, 0, 3)'$, $\mathbf{d}_2 = (0, 1, -1)'$, $\mathbf{d}_3 = (-3, 1, 2)'$,

$$D'D = \begin{pmatrix} 18 & -3 & 15 \\ -3 & 2 & -1 \\ 15 & -1 & 14 \end{pmatrix}, \quad S = \frac{D'D}{n-1} = \frac{D'D}{2} = \begin{pmatrix} 9 & -\frac{3}{2} & \frac{15}{2} \\ -\frac{3}{2} & 1 & -\frac{1}{2} \\ \frac{15}{2} & -\frac{1}{2} & 7 \end{pmatrix}.$$

Because D has rank $2 < 3$, $D'D$ is singular ($\det(D'D) = 0$), and since S is a scalar multiple of $D'D$, $\det(S) = 0$ as well. Direct check:

$$\det(D'D) = 18(28 - 1) + 3(-42 + 15) + 15(3 - 30) = 486 - 81 - 405 = 0 \implies |S| = 0.$$

Geometric interpretation: the generalized sample variance $|S|$ is proportional to the squared volume of the parallelotope spanned by the deviation vectors $\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3$ in $\mathbb{R}^n$. Since $|S| = 0$, this "volume" is zero — the three deviation vectors are linearly dependent and so lie in a plane (a 2-dimensional subspace) rather than spanning all of $\mathbb{R}^3$. The data cloud is degenerate: it has no spread in one particular direction.

(c) Total sample variance

The total sample variance is $\text{tr}(S) = S_{11} + S_{22} + S_{33} = 9 + 1 + 7 = 17$.

Problem 3.9 (8 points)

Given $X = \begin{pmatrix} 12 & 17 & 29 \\ 18 & 20 & 38 \\ 14 & 16 & 30 \\ 20 & 18 & 38 \\ 16 & 19 & 35 \end{pmatrix}$, $n = 5$, $p = 3$, with $x_3 = x_1 + x_2$ for every row.

Means: $\bar{x}_1 = \frac{12+18+14+20+16}{5} = \frac{80}{5} = 16$, $\bar{x}_2 = \frac{17+20+16+18+19}{5} = \frac{90}{5} = 18$, $\bar{x}_3 = \frac{29+38+30+38+35}{5} = \frac{170}{5} = 34 = \bar{x}_1 + \bar{x}_2$.

(a) Mean-corrected data matrix and linear dependence

$$D = X - \mathbf{1}\bar{\mathbf{x}}' = \begin{pmatrix} -4 & -1 & -5 \\ 2 & 2 & 4 \\ -2 & -2 & -4 \\ 4 & 0 & 4 \\ 0 & 1 & 1 \end{pmatrix}.$$

Column-by-column, $d_{i3} = d_{i1} + d_{i2}$ for every row i (e.g. row 1: $-4 + (-1) = -5$; row 2: $2 + 2 = 4$; etc.), so the third column of D equals the sum of the first two. This gives the dependency vector

$$\mathbf{a}' = [a_1, a_2, a_3] = [1, 1, -1], \quad \text{such that } D\mathbf{a} = \mathbf{0}.$$

(b) Sample covariance matrix, zero generalized variance, and $S\mathbf{a} = \mathbf{0}$

With columns $\mathbf{d}_1 = (-4, 2, -2, 4, 0)'$, $\mathbf{d}_2 = (-1, 2, -2, 0, 1)'$, $\mathbf{d}_3 = (-5, 4, -4, 4, 1)'$,

$$D'D = \begin{pmatrix} 40 & 12 & 52 \\ 12 & 10 & 22 \\ 52 & 22 & 74 \end{pmatrix}, \quad S = \frac{D'D}{n-1} = \frac{D'D}{4} = \begin{pmatrix} 10 & 3 & 13 \\ 3 & \frac{5}{2} & \frac{11}{2} \\ 13 & \frac{11}{2} & \frac{37}{2} \end{pmatrix}.$$

Since column 3 of D is a linear combination of columns 1 and 2 ($\mathbf{d}_1 + \mathbf{d}_2 - \mathbf{d}_3 = \mathbf{0}$), $\text{rank}(D) \leq 2 < 3$, so $D'D$ (and hence S) is singular: $|S| = 0$, i.e. the generalized sample variance is zero.

Now verify $S\mathbf{a} = \mathbf{0}$ for $\mathbf{a} = (1, 1, -1)'$:

$$S\mathbf{a} = \begin{pmatrix} 10(1) + 3(1) + 13(-1) \\ 3(1) + \frac{5}{2}(1) + \frac{11}{2}(-1) \\ 13(1) + \frac{11}{2}(1) + \frac{37}{2}(-1) \end{pmatrix} = \begin{pmatrix} 10 + 3 - 13 \\ 3 + 2.5 - 5.5 \\ 13 + 5.5 - 18.5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Since $S\mathbf{a} = \mathbf{0} = 0 \cdot \mathbf{a}$, the vector $\mathbf{a}$ (rescaled to unit length if desired) is an eigenvector of S corresponding to eigenvalue 0.

(c) Verifying $x_3 = x_1 + x_2$

Checking the original data row by row: $29 = 12 + 17$, $38 = 18 + 20$, $30 = 14 + 16$, $38 = 20 + 18$, $35 = 16 + 19$ — all hold exactly. Hence $x_3 = x_1 + x_2$ for every observation, i.e. $X\mathbf{a} = \mathbf{0}$ with $a_1 = 1, a_2 = 1, a_3 = -1$, confirming the exact linear dependence among the three columns.

Problem 3.18 (8 points)

Given $\bar{\mathbf{x}} = \begin{pmatrix} 0.766 \\ 0.508 \\ 0.438 \\ 0.161 \end{pmatrix}$, $S = \begin{pmatrix} 0.856 & 0.635 & 0.173 & 0.096 \\ 0.635 & 0.568 & 0.127 & 0.067 \\ 0.173 & 0.127 & 0.171 & 0.039 \\ 0.096 & 0.067 & 0.039 & 0.043 \end{pmatrix}$ (symmetric, so $S_{23} = S_{32} = 0.127$).

(a) Mean and variance of total energy consumption

Total consumption is $T = x_1 + x_2 + x_3 + x_4 = \mathbf{c}'\mathbf{x}$ with $\mathbf{c} = (1, 1, 1, 1)'$.

$$\text{Mean: } \mathbf{c}'\bar{\mathbf{x}} = 0.766 + 0.508 + 0.438 + 0.161 = 1.873.$$

$$\text{Variance: } \mathbf{c}'S\mathbf{c} = (\text{sum of all entries of } S) = 1.760 + 1.397 + 0.510 + 0.245 = 3.912,$$

where the row sums are $0.856 + 0.635 + 0.173 + 0.096 = 1.760$, $0.635 + 0.568 + 0.127 + 0.067 = 1.397$, $0.173 + 0.127 + 0.171 + 0.039 = 0.510$, and $0.096 + 0.067 + 0.039 + 0.043 = 0.245$.

(b) Mean, variance of the petroleum–gas excess, and its covariance with the total

Let $D = x_1 - x_2 = \mathbf{b}'\mathbf{x}$ with $\mathbf{b} = (1, -1, 0, 0)'$.

$$\text{Mean: } \mathbf{b}'\bar{\mathbf{x}} = 0.766 - 0.508 = 0.258.$$

$$\text{Variance: } \mathbf{b}'S\mathbf{b} = S_{11} + S_{22} - 2S_{12} = 0.856 + 0.568 - 2(0.635) = 1.424 - 1.270 = 0.154.$$

$$\text{Covariance with } T: \mathbf{b}'S\mathbf{c} = (S_{11} + S_{12} + S_{13} + S_{14}) - (S_{12} + S_{22} + S_{23} + S_{24}) = 1.760 - 1.397 = 0.363.$$

Problem 3.20 (8 points)

Snow data, $n = 25$ storms, $x_1 =$ duration (hours), $x_2 =$ clearing effort (hours).

(a) Mean and variance of $x_2 - x_1$ via summary statistics

Summing the 25 observations: $\sum x_1 = 235.5$, $\sum x_2 = 481.8$, $\sum x_1^2 = 2557.75$, $\sum x_2^2 = 10778.98$, $\sum x_1 x_2 = 4861.90$.

$$\bar{x}_1 = \frac{235.5}{25} = 9.42, \quad \bar{x}_2 = \frac{481.8}{25} = 19.272.$$

Corrected sums of squares and cross products:

$$S_{xx} = \sum x_1^2 - n\bar{x}_1^2 = 2557.75 - 25(9.42)^2 = 2557.75 - 2218.41 = 339.34,$$

$$S_{yy} = \sum x_2^2 - n\bar{x}_2^2 = 10778.98 - 25(19.272)^2 = 10778.98 - 9285.2496 = 1493.7304,$$

$$S_{xy} = \sum x_1 x_2 - n\bar{x}_1 \bar{x}_2 = 4861.90 - 25(9.42)(19.272) = 4861.90 - 4538.556 = 323.344.$$

Sample variances/covariance (divisor $n - 1 = 24$):

$$s_1^2 = \frac{339.34}{24} = 14.1392, \quad s_2^2 = \frac{1493.7304}{24} = 62.2388, \quad s_{12} = \frac{323.344}{24} = 13.4727.$$

Let $D = x_2 - x_1$. Then

$$\bar{D} = \bar{x}_2 - \bar{x}_1 = 19.272 - 9.42 = 9.852,$$

$$s_D^2 = s_2^2 + s_1^2 - 2s_{12} = 62.2388 + 14.1392 - 2(13.4727) = 76.3780 - 26.9453 = 49.4326.$$

(b) Mean and variance of $x_2 - x_1$ via individual differences

Computing $d_j = x_{2j} - x_{1j}$ for each of the 25 storms and summing directly: $\sum d_j = 246.3$, so $\bar{d} = \frac{246.3}{25} = 9.852$ — identical to part (a), as it must be, since the mean of a difference equals the difference of the means for any data set.

For the variance, $\sum d_j^2 = 3612.93$, so

$$S_{dd} = \sum d_j^2 - n\bar{d}^2 = 3612.93 - 25(9.852)^2 = 3612.93 - 2426.5476 = 1186.3824,$$

$$s_d^2 = \frac{S_{dd}}{n - 1} = \frac{1186.3824}{24} = 49.4326.$$

This matches the value obtained in (a) exactly. **Comparison:** the two approaches agree perfectly ($\bar{d} = 9.852$, $s_d^2 \approx 49.4326$ both ways), confirming that computing summary statistics on the raw variables and combining them algebraically ($s_2^2 + s_1^2 - 2s_{12}$) is equivalent to forming the differences first and then computing their mean and variance directly — the variance-of-a-linear-combination formula is exact, not an approximation.